

Practicum No.4**Date: 24/07/26****Cumulative Distribution Evaluation of Machine Learning Features Using Random Variables****I. Practical Significance.**

Cumulative Distribution Analysis plays a fundamental role in understanding machine learning features by interpreting them as random variables governed by probability distributions. By computing the CDF $P(X \leq x)$, this practicum enables learners to analyze the probability structure, spread, skewness, and concentration of feature values within a dataset. Unlike simple descriptive statistics such as mean and variance, the CDF provides a complete probabilistic view of how data accumulates across its range, which is essential for threshold-based decision-making, anomaly detection, and feature scaling. In predictive modelling, CDF analysis helps identify outliers, assess distribution stability, and compare feature behavior across different classes or populations. Furthermore, it strengthens conceptual understanding of random variables, empirical probability, and statistical inference—core mathematical foundations underlying machine learning algorithms. Thus, this practicum bridges probability theory with real-world data analytics, enhancing interpretability and reliability in model development.

II Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency

- Use the principles of Probability and Statistics to solve problems in Computer Science. (As stated in the Course Structure)
- a) Develop Python program to solve problems related to Computer Science. (CDO)
- b) Use memory management techniques effectively. (CDO)
- c) Interpret the observations and results. (CDO)
- d) Follow professional ethics (ADO)

III Relevant CO(s) (As stated in the Course Structure)

- CO2: Model the data that has uncertainties.

IV Practical Outcome (PrO) (As stated in the Course Structure)

- Compute the cumulative probability function of the given machine learning dataset using random variables.

V Related ADO(s) (As stated in the Course Structure)

- a) Follow professional ethics

VI. Minimum Underpinning Theory (Include relevant content & images for this practicum.)

Computing the Cumulative Distribution Function (CDF) of a machine learning dataset is fundamentally grounded in the theory of random variables and probability distributions.

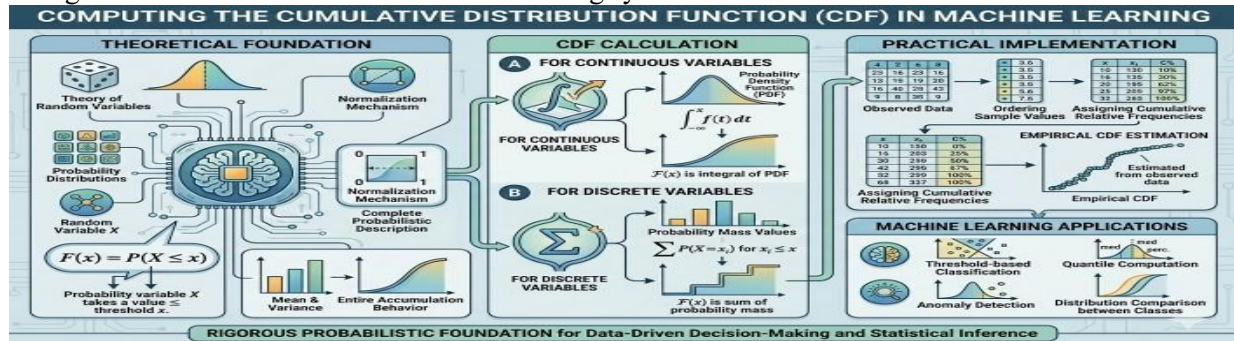
If a feature is modeled as a random variable X , its cumulative distribution function is defined as

$F(x) = P(X \leq x)$, which represents the probability that the variable takes a value less than or equal to a specified threshold.

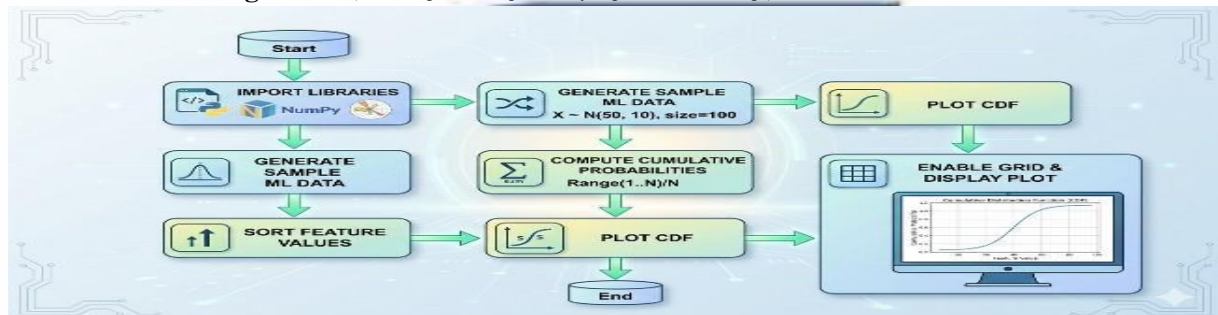
For continuous variables, the CDF is obtained by integrating the probability density function (PDF), while for discrete variables it is computed as a summation of probability mass values.

In practical machine learning datasets, the empirical CDF is estimated from observed data by ordering the sample values and assigning cumulative relative frequencies, thereby approximating the theoretical distribution. The CDF acts as a normalization mechanism, ensuring probabilities remain bounded between 0 and 1 and providing a complete probabilistic description of the feature. Unlike mean or variance, which summarize central tendency or spread, the CDF captures the entire accumulation behavior of data across its domain. Theoretically, it supports threshold-based classification, quantile computation, anomaly detection, and distribution comparison between classes. By interpreting features as realizations of random variables,

cumulative distribution analysis establishes a rigorous probabilistic foundation for data-driven decision-making and statistical inference in machine learning systems.



VII. Practicum Algorithm *(Include probable photos of experimental set-up.)*



VIII. Resources Required

S. No.	Resource	Suggested Broad Specification	Quantity
1	Laptop	Minimum Intel 5 Processor	1 No.
2	Compiler	PYTHON NumPy: Python library for numerical computing, providing support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these structures efficiently.	1 No.

IX. Safety Precautions

NA

X. Procedure *(Self-Instructional)*

- Input** Feature Data: Select a specific numerical feature (random variable X) from your machine learning dataset.
- Sort** the Feature Values: Arrange all observed values in the dataset in ascending order (from smallest to largest). Let the sorted data be $x_1, x_2, \dots, x_n$.
- Assign** Rank-Based Probabilities: For each sorted value, calculate its cumulative probability by dividing its position index (i) by the total number of observations (n). Formula: $P(X \cdot x_i) = i/n$.
- Define** the Cumulative Function: Map each feature value to its calculated probability. This creates a function $F(x)$ that always stays between 0 and 1.
- Compare** Values to Thresholds (Analysis): Evaluate the probability of data falling below a specific value: If $F(x)$ is near 0.5. The value is near the median of the feature. If $F(x)$ is near 1.0. The value is an upper outlier or maximum.
- Display** Results: Plot the results as a curve with Feature Values on the x-axis and Cumulative Probability on the y-axis. Conclude with a summary of the data distribution (e.g., identifying the 95th percentile).

Pseudocode

Start

- Input:** A feature vector X containing n numerical observations.

2: **Sort:** X_{sorted} Sort the values of X in ascending order.

3: **Initialize:** Create an empty list or array CDF_values .

4: **Compute:** For each index i from 1 to n in X_{sorted} : $P_i \leftarrow i/n$ (Calculate relative cumulative frequency). Add P_i to CDF_values End For

5: **Analyze Probability:** If P_i for a given x is 0.5 then Output " x is the Median of the dataset" Else If P_i is 0.95 then Output " x is the 95th percentile (Upper Threshold)".

6: **Display:** Output the mapping of X_{sorted} to CDF_values .

7: **Plot** X_{sorted} (x-axis) vs CDF_values (y-axis). End

XI. Actual Resources Used (To be filled by the students)

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop/Computer	Windows 11, Intel Processor	1	Used for program development	
2.	Python	Python 3.x	1	Programming language	
3.	Python IDE / Google Colab	Python execution environment	1	Used to run the programs	
4.	NumPy Library	Python library for numerical computing	1	Used for numerical operations if required	

XII. Actual Procedure Followed (To be filled by the students)

- Opened the Python programming environment (e.g., Google Colab / Python IDE).
- Imported the required library (NumPy) for numerical and array operations.
- Defined the discrete random variable values X and their corresponding probability mass function $P(X)$ for each scenario (CPU-intensive requests, lost packets, and suspicious login attempts).
- Calculated the Cumulative Distribution Function (CDF) using NumPy's cumulative sum function (`np.cumsum(P)`).
- Iterated through the values to format and display the tabular representation of X , $P(X)$, and CDF $F(X)$.
- Extracted specific cumulative probabilities (such as $F(1)$, $F(2)$, and $F(3)$) corresponding to targeted thresholds.
- Executed the scripts, captured console outputs/screenshots, and documented observations and interpretations.
- Executed the Python program and displayed the results.
- Recorded and interpreted the obtained cumulative probabilities.

XIII. Observations (To be filled by the students)

S.No.	Input	Output	Remarks
1	CPU-intensive requests: $X = [0, 1, 2, 3, 4]$, $P = [0.10, 0.20, 0.35, 0.25, 0.10]$	$F(1) = 0.30$, $F(3) = 0.90$	90% chance of receiving ≤ 3 CPU-intensive requests

2	Lost packets: $X = [0, 1, 2, 3, 4]$, $P = [0.15, 0.25, 0.30, 0.20, 0.10]$	$F(2) = 0.70$, $F(3) = 0.90$	70% probability of losing ≤ 2 packets; 90% probability of losing ≤ 3 packets
3	Suspicious login attempts: $X = [0, 1, 2, 3, 4]$, $P = [0.15, 0.25, 0.30, 0.20, 0.10]$	$F(2) = 0.70$, $F(3) = 0.90$	Cumulative risk stays at 70% for ≤ 2 attempts and reaches 90% for ≤ 3 attempts

XIV. Results/Observations *(To be filled by the students)*

1.

```
import numpy as np
```

```
# Number of CPU-intensive requests
```

```
X = np.array([0, 1, 2, 3, 4])
```

```
# Probability P(X)
```

```
P = np.array([0.10, 0.20, 0.35, 0.25, 0.10])
```

```
# Compute CDF
```

```
CDF = np.cumsum(P)
```

```
print("X\tP(X)\tCDF F(X)")
```

```
for x, p, cdf in zip(X, P, CDF):
```

```
print(f'{x}\t{p:.2f}\t{cdf:.2f}')
```

```
# Required CDF values
```

```
print("\nF(1) =", CDF[X == 1][0])
```

```
print("F(3) =", CDF[X == 3][0])
```

X	P(X)	CDF F(X)
0	0.10	0.10
1	0.20	0.30
2	0.35	0.65
3	0.25	0.90
4	0.10	1.00

$F(1) = 0.30000000000000004$
 $F(3) = 0.9$

2.

```

import numpy as np

# Number of lost packets
X = np.array([0, 1, 2, 3, 4])

# Probability P(X)
P = np.array([0.15, 0.25, 0.30, 0.20, 0.10])

# Compute CDF
CDF = np.cumsum(P)

# Display PMF and CDF
print("X\tP(X)\tCDF F(X)")
print("-----")
for x, p, f in zip(X, P, CDF):
    print(f"{x}\t{p:.2f}\t{f:.2f}")

# Required CDF values
print("\nF(2) =", CDF[X == 2][0])
print("F(3) =", CDF[X == 3][0])

```



```

lenwin@fedora: /run/media/lenwin/New Volume/A
R/Maths/Prac Questions/p4b.py"
X      P(X)    CDF F(X)
-----
0      0.15    0.15
1      0.25    0.40
2      0.30    0.70
3      0.20    0.90
4      0.10    1.00

F(2) = 0.7
F(3) = 0.8999999999999999
lenwin@fedora: /run/media/lenwin/New Volume/A

```

3.

```
import numpy as np
```

```
# Number of suspicious login attempts
```

```
X = np.array([0, 1, 2, 3, 4])
```

```
# Probability distribution
```

```
P = np.array([0.15, 0.25, 0.30, 0.20, 0.10])
```

```
# Compute CDF
```

```
CDF = np.cumsum(P)
```

```
# Display the probability distribution and CDF
```

```
print("X\tP(X)\tF(X)")
```

```
for x, p, f in zip(X, P, CDF):
```

```
print(f"{x}\t{p:.2f}\t{f:.2f}")
```

```
# Find required CDF values
```

```
print("\nF(2) =", CDF[2])
```

```
print("F(3) =", CDF[3])
```

```

F(3) = 0.8999999999999999
lenwin@fedora:/run/media/lenwin/New Volume/A
R/Maths/Prac Questions/p4c.py"
X      P(X)    F(X)
0      0.15    0.15
1      0.25    0.40
2      0.30    0.70
3      0.20    0.90
4      0.10    1.00

F(2) = 0.7
F(3) = 0.8999999999999999
lenwin@fedora:/run/media/lenwin/New Volume/A

```

XV. Interpretation of Results *(Students to state the meaning of the above-obtained results/observations)*

The results demonstrate the evaluation of the **Cumulative Distribution Function (CDF)** across three machine learning scenarios modeled via discrete random variables: CPU-intensive requests, lost network packets, and suspicious login attempts. For CPU-intensive requests, the cumulative probability $F(3) = 0.90$ indicates a 90% probability that 3 or fewer requests arrive. For packet losses and suspicious login attempts, $F(2) = 0.70$ and $F(3) = 0.90$ show that 70% and 90% of observed occurrences fall at or below 2 and 3 events, respectively. CDF analysis effectively captures data accumulation for setting system thresholds, capacity management, and security alerting.

XVI. Conclusions *(Students to draw conclusions (or take decisions) based on the interpretation of results)*

Thus, the Python programs were successfully executed using NumPy to calculate and visualize the **Cumulative Distribution Function (CDF)** across CPU request loads, packet loss events, and suspicious login security features. The cumulative probabilities verified how probability accumulates across feature thresholds, establishing a quantitative foundation for risk profiling, anomaly detection, and decision-making in machine learning environments.

XVII. Suggested Practicum Related Questions:

Note: Below are a few questions related to industry-specific higher-order thinking skills (HOT) that teachers must use to ensure students achieve the predetermined course outcomes..

1. **Contrast** the detection of sensor drift using a Cumulative Distribution Function versus standard central tendency measures.
2. **Interpret** the effectiveness of CDF-based normalization (Quantile Transformation) when processing highly skewed industrial datasets.
3. **Interpret** the physical implications of a persistent horizontal plateau within a sensor's empirical CDF plot.
4. **Differentiate** between the insights provided by a Cumulative Distribution Function and a Probability Density Function when characterizing the tail-end risks of a manufacturing process.
5. **Derive** critical safety thresholds by identifying the specific feature value where the failure probability $F(x)$ reaches a critical risk limit.
6. **Analyze** the impact of sensor quantization noise on the smoothness and predictive reliability of the resulting empirical CDF.
7. **Predict** the specific geometric deformation of a vibration CDF curve as a mechanical component approaches its end-of-life.
8. **Appraise** the computational efficiency of using $F(b) - F(a)$ for tolerance intervals compared to integrating a Probability Density Function.
9. **Formulate** a dynamic alarm strategy that utilizes the 99th percentile of a shifting CDF to accommodate seasonal environmental variables.
10. **Critique** the assumption of distribution stationarity when deploying machine learning models in evolving industrial environments.

XVIII References / Suggestions for further reading

1. **Probability and Statistics for Machine Learning: A Textbook (Charu C. Aggarwal)**
(This comprehensive resource connects fundamental probability theory directly to machine learning applications.)
2. **Probability and Distributions – Mathematics for Machine Learning**
<https://www.youtube.com/watch?v=ua-JYs6Eipl>
3. **Introduction to Random Variables** <https://www.youtube.com/watch?v=3v9w79Nhsf>

XIX Suggested Assessment Scheme

The performance indicators given serve as a guideline for assessing the 'process-related skills/LOs' (marks to be awarded in real-time in the laboratory by the faculty member, as these cannot be measured after the practicum is over) and 'product-related skills/LOs'

Note: The weightage for the process-related skills and product-related skills will change depending on the PrO, COs, and the competency related to the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related Skills: 18 Marks - 60%			
1	Selection of relevant function	6	
2	Sequence of Computations	10	
3	Ethics followed	2	

Product-related Skills: 12 Marks - 40%			
1	Output Accuracy	4	
2	Interpretation of results	2	
3	Conclusions	2	
4	Practical related questions	2	
5	Submitting the Journal on time	2	
Total		30	

To be filled by Student:

S.No	Name of the Student	Register No.
1	LESLIY ROBINSON A	URK25CS1102

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Date of Evaluation
Process-Related Skills (18)	Product-Related Skills (12)	Total (30)		

Karunya